PROBABILITY DISTRIBUTIONS

Unified Quick Reference

Mathematical Visualisation & Engineering Applications Edition

CHEATSHEET 09

V • G • R

V • GUIDANCE • RESEARCH

1 BASIC PROBABILITY CONCEPTS

Sample space S = all outcomes; event $A \subseteq S$.

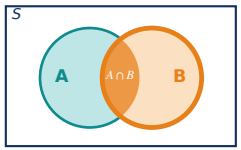
Axioms: $P(A) \geq 0$; $P(S) = 1$; $P(\cup A_i) = \sum P(A_i)$ **(disjoint).**

Complement

$$P(A^c) = 1 - P(A)$$

Addition

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



P(A|B): restrict to B, take the share inside A.

Conditional

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplication

$$P(A \cap B) = P(A|B)\,P(B)$$

Bayes' theorem

$$P(A_i|B) = \frac{P(B|A_i)\,P(A_i)}{\sum_j P(B|A_j)\,P(A_j)}$$

A_j: partition of S. Ex.: M_1 (60%, 2% defective), M_2 (40%, 5%):

$P(M_1|D) = \frac{0.6 \times 0.02}{0.6 \times 0.02 + 0.4 \times 0.05} = 0.375$

2 RANDOM VARIABLES: PMF, PDF, CDF

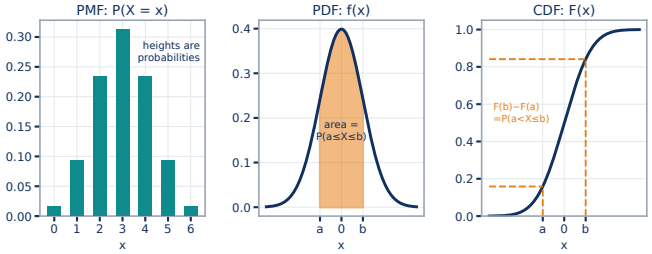
- Discrete: countable values; PMF $p(x) = P(X = x)$.
- Continuous: interval values; PDF $f(x)$, and $P(X = x) = 0$.
- CDF $F(x) = P(X \leq x)$ is non-decreasing, from 0 to 1.

Discrete: $\sum_x P(X = x) = 1$

Continuous: $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$F(x) = P(X \leq x)$

$P(a \leq X \leq b) = \int_a^b f(x)\,dx$



PMF: $P(X = x)$ itself. PDF: only areas are probabilities.

Plots: Binomial(6, 0.5) PMF; $N(0, 1)$ PDF and CDF.

3 EXPECTATION AND VARIANCE

Discrete

$$E(X) = \sum_x x\,P(X = x)$$

Continuous

$$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$$

Variance

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Linear change

$E(aX + b) = aE(X) + b$

$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$

Sums

$E(X + Y) = E(X) + E(Y)$

$\mathrm{Var}(X + Y) = \mathrm{Var}X + \mathrm{Var}Y$

E is always additive; Var adds only for independent X, Y.

Fair die: $E = 3.5$, $\mathrm{Var} = 35/12$. $\sigma = \sqrt{\mathrm{Var}}$ has the units of X.

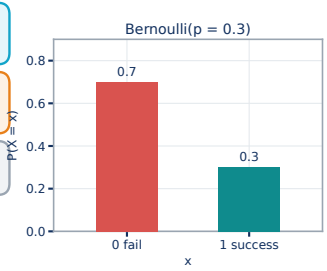
4 BERNOULLI DISTRIBUTION

$X \sim \mathrm{Bernoulli}(p)$

$P(X = x) = p^x(1-p)^{1-x},\ x \in \{0, 1\}$

$E(X) = p$ | $\mathrm{Var} = p(1-p)$

Use: success/failure, component reliability, binary classification.



5 BINOMIAL DISTRIBUTION

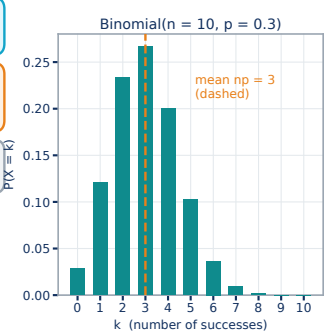
$X \sim \mathrm{Binomial}(n, p)$

$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$

$E(X) = np$ | $\mathrm{Var} = np(1-p)$

Conditions (all four):

- fixed number n of trials
- two outcomes per trial
- same p each trial
- trials independent



Worked example (quality control)

Defect rate $p = 0.1$, sample $n = 10$. Find $P(X \leq 1)$.

$P(X \leq 1) = 0.9^{10} + 10(0.1)(0.9)^9 = 0.3487 + 0.3874 = 0.7361$

Mean defects $np = 1$; $\sigma = \sqrt{0.9} \approx 0.949$.

Use: quality control, defective parts, reliability tests.

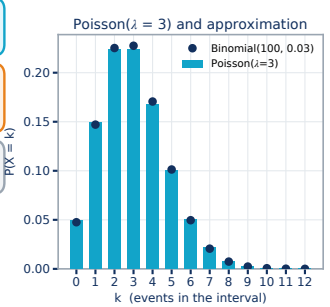
6 POISSON DISTRIBUTION

$X \sim \mathrm{Poisson}(\lambda)$

$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!},\ k = 0, 1, \ldots$

$E(X) = \lambda$ | $\mathrm{Var} = \lambda$

Rare-event counts in a fixed interval (time, length, area); λ = mean count per interval.



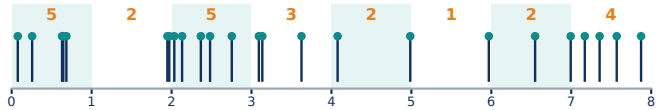
Binomial → Poisson: n large, p small, $\lambda = np$ fixed

(rule of thumb $n \geq 20$, $p \leq 0.05$). Dots: Bin(100, 0.03).

Ex.: $\lambda = 3$ errors/min: $P(X = 2) = e^{-3}3^2/2! = 0.2240$.

Use: arrivals, communication errors, machine failures.

Rare events on a time line ($\lambda = 3$ per unit interval): counts per interval vary around 3



$P(X = 0..4)$ for $\lambda = 3$: 0.0498, 0.1494, 0.2240, 0.2240, 0.1680

7 GEOMETRIC DISTRIBUTION

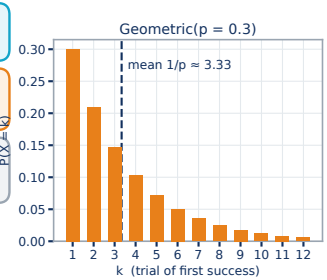
$X \sim \mathrm{Geometric}(p)$

$P(X = k) = (1-p)^{k-1}p,\ k = 1, 2, \ldots$

$E(X) = \frac{1}{p}$ | $\mathrm{Var} = \frac{1-p}{p^2}$

Trials needed for the first success.

Ex.: $p = 0.3$: $P(X = 3) = 0.7^2(0.3) = 0.147$.



Convention: counting failures instead ($k = 0, 1, \ldots$) gives $P = (1-p)^k p$ and mean $(1-p)/p$. State which you use.

8 UNIFORM DISTRIBUTION

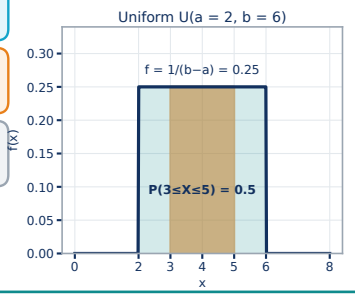
$X \sim U(a, b)$

$f(x) = \frac{1}{b-a},\ a \le x \le b$

$E = \frac{a+b}{2}$ $\quad \text{Var} = \frac{(b-a)^2}{12}$

f = 0 outside [a, b]. Equal likelihood on equal-length subintervals.

Ex.: U(2,6): mean 4, Var 4/3.



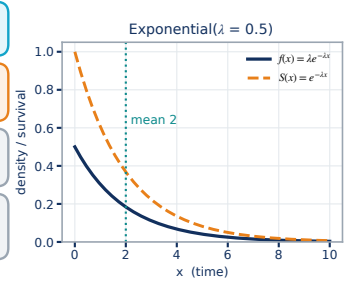
9 EXPONENTIAL DISTRIBUTION

$X \sim \text{Exp}(\lambda)$

$f(x) = \lambda e^{-\lambda x},\ x \ge 0$

$F(x) = 1 - e^{-\lambda x},\ S(x) = e^{-\lambda x}$

$E = \frac{1}{\lambda}$ $\quad \text{Var} = \frac{1}{\lambda^2}$



Memoryless property

$$P(X > s + t \mid X > s) = P(X > t)$$

Ex.: $\lambda = 0.5$**/h (mean 2 h):** $P(X > 3) = e^{-1.5} = 0.2231$.

Use: reliability, waiting times, lifetime models. Failure rate = λ.

12 MOMENT-GENERATING FUNCTION

$M_X(t) = E(e^{tX})$ $\quad = \sum_x e^{tx} P(X = x)$ or $\int e^{tx} f(x)\,dx$

$E(X) = M_X'(0)$ $\quad E(X^2) = M_X''(0)$ $\quad \text{Var} = M''(0) - [M'(0)]^2$

Moments by differentiation; equal MGFs (finite near 0) identify the same distribution.

Bernoulli	$1 - p + pe^t$	**Binomial**	$(1 - p + pe^t)^n$
Poisson	$e^{\lambda(e^t - 1)}$	**Geometric**	$\frac{pe^t}{1-(1-p)e^t}$
Exponential	$\frac{\lambda}{\lambda - t},\ t < \lambda$	**Normal**	$e^{\mu t + \sigma^2 t^2/2}$
Gamma	$(1 - \theta t)^{-\alpha},\ t < 1/\theta$		

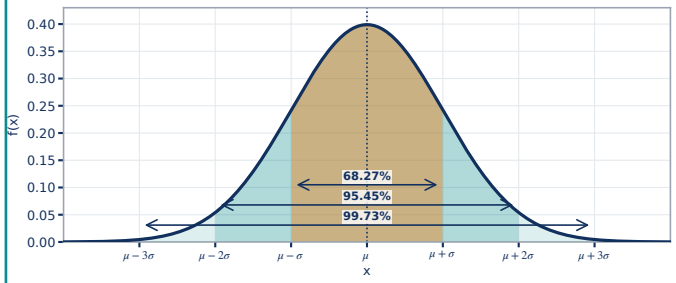
Uniform: $M(t) = \frac{e^{bt} - e^{at}}{(b-a)t}$ $(t \ne 0)$.

10 NORMAL DISTRIBUTION

$X \sim N(\mu, \sigma^2)$ $\quad E = \mu,\ \text{Var} = \sigma^2$

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$

$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$ $\quad P(X \le x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$



Empirical rule: $P(|X - \mu| < k\sigma) = 0.6827,\ 0.9545,\ 0.9973$ **for** $k = 1, 2, 3$.

Ex.: $N(50, 2^2)$: $P(48 \le X \le 54) = \Phi(2) - \Phi(-1) = 0.8186$.

11 GAMMA AND BETA DISTRIBUTIONS

$X \sim \text{Gamma}(\alpha, \theta),\ x > 0$

$f = \frac{x^{\alpha-1} e^{-x/\theta}}{\Gamma(\alpha)\theta^\alpha}$

$E = \alpha\theta,\ \text{Var} = \alpha\theta^2$

α: shape, θ: scale.

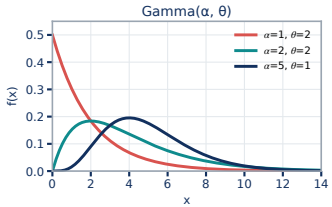
Gamma(1, θ) = Exp(rate $1/\theta$).

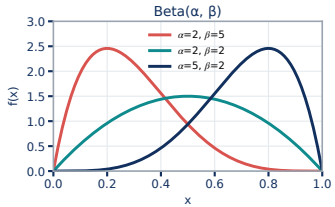
$X \sim \text{Beta}(\alpha, \beta),\ 0 < x < 1$

$f = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$

$E = \frac{\alpha}{\alpha+\beta}$

$\text{Var} = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$





Gamma: waiting times, reliability. Beta: proportions, Bayesian inference (Beta(1, 1) = U(0, 1)).

16 COMMON ERRORS

- ✗ Confusing PMF with PDF
- ✗ Binomial without independence
- ✗ Confusing σ with σ^2
- ✗ Omitting support restrictions
- ✗ Normal approx. unchecked ($np \ge 5$)
- ✗ Forgetting normalisation (total 1)
- ✗ Mixing λ, mean and variance (Poisson: both $= \lambda$)
- ✗ Wrong geometric convention
- ✗ Treating a PDF value as probability

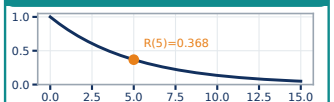
13 COMPARATIVE DISTRIBUTION TABLE

Distribution	Params	Mean	Variance	Typical use
Bernoulli	p	p	$p(1-p)$	binary outcome
Binomial	n, p	np	$np(1-p)$	defect counts
Poisson	λ	λ	λ	arrivals, errors
Geometric	p	$1/p$	$(1-p)/p^2$	trials to success
Uniform	a, b	$(a+b)/2$	$(b-a)^2/12$	random phase
Exponential	λ	$1/\lambda$	$1/\lambda^2$	lifetimes, waits
Normal	μ, σ^2	μ	σ^2	tolerances, noise
Gamma	α, θ	$\alpha\theta$	$\alpha\theta^2$	sum of waits
Beta	α, β	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	proportions, Bayes

Gamma uses (shape α, scale θ); Geometric counts trials, $k \ge 1$.

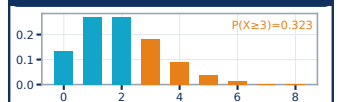
14 ENGINEERING APPLICATIONS

A Reliability



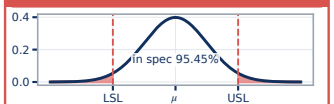
$R(t) = e^{-\lambda t}$, $\lambda = 0.2$/h

B Communication



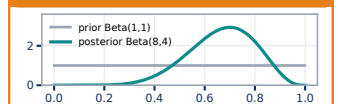
errors/block ~ Poisson(2)

C Manufacturing



spec $\mu \pm 2\sigma$: 4.55% out

D Data science



7/10 successes: Beta(1,1) → Beta(8,4)

15 QUICK METHOD-SELECTION GUIDE

Random variable X: what is observed?

Discrete (counts)		Continuous (measurements)	
Two outcomes, one trial	Bernoulli	Constant density on [a,b]	Uniform
Fixed n, independent trials	Binomial	Memoryless waiting time	Exponential
Counts in an interval	Poisson	Symmetric, bell-shaped	Normal
Trials to first success	Geometric	Positive waiting time, shape α	Gamma
		Proportion in (0, 1)	Beta

Memoryless: Exponential (continuous), Geometric (discrete).

Counts with mean ≈ variance suggest Poisson; variance < mean suggests Binomial.